# THE SUPER-RICCATI SHADOW TOWER AND THE VERDIER COMPLEMENTARITY PROBLEM

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ABSTRACT. The super-Riccati tower gives a theorem for shadow coefficients attached to a homogeneous primary line after the line-restricted Maurer–Cartan equation is put in the Riccati normalisation fixed below. The seed data $S_3$, $S_4$ are external OPE data; the recurrence starts at $r = 5$. The parity of the line first changes the diagonal $(4, 4)$-term in $S_6$. The Verdier complementarity statement for $Y(\mathfrak{sl}(m|n))$ is separate from the recurrence. The rank-two calculation in the stated super-trace normalisation gives $3 = \dim \mathfrak{sl}(2)$. Its conjectural family extension has constant $\dim \mathfrak{sl}(M) = M^2 - 1$, where $M = \max(m, n)$. The uniform scalar convention with numerator $M$ instead has constant $M$. The Berezinian normalisation has the distinct conjectural constant $|m - n|$ only after the Berezinian–Verdier datum is fixed. Shadow coefficients and Igusa denominator scalars are coefficient data; they do not construct compact Hall objects, quasi-NCCR realisations, Hall–Drinfeld doubles, orientation systems, PBW bases, or compact CoHA.

## 1. RICCATI RECURRENCE

**Definition 1** (Riccati-normalised primary line). Fix a homogeneous primary line $\ell$ with parity $|\ell|$ and nonzero coefficient $\kappa_\ell$. Let $S_r$ be the shadow coefficient of degree $r$, and choose line descendants $\theta_r$ with

$$|\theta_r| = (r-1)|\ell| \pmod 2.$$

Set

$$f(j,k) = \begin{cases} 1, & j < k, \\ \frac{1}{2}, & j = k. \end{cases}$$

The line is Riccati-normalised if the degree-$r$ component ($r \geq 5$) of the line-restricted Maurer–Cartan equation is

$$2r\kappa_\ell S_r + \sum_{\substack{j+k=r+2 \\ 3\leq j\leq k<r}} (-1)^{(j-1)(k-1)|\ell|} f(j,k)\, j\, k\, S_j S_k = 0.$$

Here $f(j,k)$ is the unordered Wick-pair factor obtained after the ordered quadratic term $\frac{1}{2}[\Theta_\ell, \Theta_\ell]$ is quotiented to $j \leq k$: off-diagonal pairs occur once, while diagonal pairs carry the residual $\frac{1}{2}$ symmetry factor in this normalisation.

**Definition 2** (Admissible seed). An admissible seed for a Riccati-normalised line $\ell$ is a pair $(S_3, S_4)$ obtained from the line-restricted OPE data, satisfying locality, parity, and non-degeneracy of the Zamolodchikov pairing $\kappa_\ell \neq 0$. These two scalars are not produced by the recurrence. If $\ell$ is a pure single-mode odd line in a parity-preserving theory, the scalar cubic coefficient satisfies $S_3 = 0$. The quartic seed $S_4$ vanishes only when the chosen line has no genuine weight-four even composite channel; otherwise it is independent input.

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**Theorem 3** (Super-Riccati recurrence). *For a Riccati-normalised primary line with admissible seed pair $(S_3, S_4)$, the coefficients for $r \geq 5$ satisfy*

$$S_r = -\frac{1}{2r\kappa_\ell} \sum_{\substack{j+k=r+2 \\ 3\leq j\leq k<r}} (-1)^{(j-1)(k-1)|\ell|} f(j,k)\, j\, k\, S_j S_k.$$

*Proof.* The Riccati normalisation says precisely that the degree-$r$ component of the line-restricted Maurer–Cartan equation is

$$2r\kappa_\ell S_r + \sum_{\substack{j+k=r+2 \\ 3\leq j\leq k<r}} (-1)^{(j-1)(k-1)|\ell|} f(j,k)\, j\, k\, S_j S_k = 0$$

with the Koszul sign computed from $|\theta_j||\theta_k| = (j-1)(k-1)|\ell|$. The coefficient $2r\kappa_\ell$ is nonzero because $\kappa_\ell \neq 0$ and the base field has characteristic zero. Solving the displayed equation for $S_r$ gives the recurrence. $\square$

*Remark* 4. On a pure odd line the cubic seed satisfies $S_3 = 0$. The quartic seed $S_4$ is non-recursive OPE data and may vanish depending on the chosen line.

**Proposition 5** (First recursive coefficients). *For every Riccati-normalised line,*

$$S_5 = -\frac{6}{5\kappa_\ell} S_3 S_4$$

*and*

$$S_6 = -\frac{1}{12\kappa_\ell}\Big(15 S_3 S_5 + 8(-1)^{|\ell|} S_4^2\Big).$$

*Equivalently, in seed-only form,*

$$S_6 = \frac{S_4}{6\kappa_\ell^2}\Big(9S_3^2 - 4(-1)^{|\ell|}\kappa_\ell S_4\Big).$$

*Thus the even-line recurrence is the bosonic Riccati recurrence, while the first parity-dependent recursive sign occurs in the diagonal $(4,4)$ contribution to $S_6$. The values through $S_6$ are normalisation fingerprints for the chosen line; they do not determine a full shadow class, a compact source, or a Verdier datum without the seed and line normalisation.*

*Proof.* For $r = 5$, the only pair is $(j,k) = (3,4)$, and $(-1)^{(2)(3)|\ell|} = 1$, so

$$S_5 = -\frac{1}{10\kappa_\ell}\, 12 S_3 S_4 = -\frac{6}{5\kappa_\ell} S_3 S_4.$$

For $r = 6$, the pairs are $(3,5)$ and $(4,4)$. The first has sign $(-1)^{(2)(4)|\ell|} = 1$ and coefficient 15. The second has coefficient $f(4,4)\cdot 16 = 8$ and sign $(-1)^{(3)(3)|\ell|} = (-1)^{|\ell|}$. Substitution into the recurrence gives the displayed formula.

Equivalently, the two recursive levels are the following term-by-term expansions:

| $r$ | $(j,k)$ | $(-1)^{(j-1)(k-1)\|\ell\|} f(j,k)\,jk$ | contribution |
|---|---|---|---|
| 5 | $(3,4)$ | 12 | $12S_3S_4$ |
| 6 | $(3,5)$ | 15 | $15S_3S_5$ |
| 6 | $(4,4)$ | $8(-1)^{\|\ell\|}$ | $8(-1)^{\|\ell\|}S_4^2$. |

Substituting the displayed value of $S_5$ gives the seed-only formula. $\square$

## 2. Shadow data and Hall obstructions

**Definition 6.** The shadow datum of the primary line is the scalar sequence

$$\mathrm{Sh}_\ell(A) = (S_2, S_3, S_4, S_5, \ldots), \qquad S_2 := \kappa_\ell, \in k^{\mathbb{N}_{\geq 2}}.$$

The obstruction tuple for a compact Hall source is

$$\mathfrak{o}_{\mathrm{Hall}} = (\mathfrak{o}_{\mathrm{or}}, \mathfrak{o}_{\mathrm{corr}}, \mathfrak{o}_{\mathrm{coprod}}, \mathfrak{o}_{\mathrm{pair}}, \mathfrak{o}_{\mathrm{rad}}, \mathfrak{o}_{\mathrm{PBW}}, \mathfrak{o}_{\mathrm{comp}}),$$

where the entries ask for an orientation/Pfaffian line, compact Hall correspondences, a coproduct, a non-degenerate Hopf pairing, radical descent, PBW/no-extra-relations compatibility, and compact support.

**Proposition 7.** *The recurrence for* $\mathrm{Sh}_\ell(A)$ *constructs no compact Hall object, orientation datum, PBW theorem, or compact CoHA:*

$$\mathrm{Sh}_\ell(A) \;\not\Longrightarrow\; \mathfrak{o}_{\mathrm{Hall}} = 0.$$

*Proof.* The map $A \mapsto \mathrm{Sh}_\ell(A)$ lands in a sequence of scalars computed from one primary line. Compact Hall data live on a moduli problem with extension correspondences and vanishing-cycle sheaves. PBW requires a filtered algebra and an associated-graded comparison. Orientation requires a square root of the determinant/Pfaffian line. None of these structures is present in the codomain $k^{\mathbb{N}_{\geq 2}}$, and no map from the displayed scalar sequence to the obstruction tuple is constructed by the Riccati recurrence. A scalar match can be a necessary compatibility check; it is not a construction of the Hall source and not a vanishing proof for $\mathfrak{o}_{\mathrm{Hall}}$. □

**Proposition 8** (Igusa scalar normalisation)**.** *When the $K3 \times E$ denominator is compared with a shadow tower, the physical unnormalised genus-two scalar, the OP square, and the OP/DT reciprocal scalar are*

$$\Phi_{10}^{\mathrm{un}} = \Delta_5^2, \qquad D_5 = 64^{-1}\Delta_5, \qquad \Phi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2,$$
$$Z_{\mathrm{OP/DT}} = -\left(\Phi_{10}^{\mathrm{OP}}\right)^{-1} = -D_5^{-2} = -4096\,\Delta_5^{-2}.$$

*The equality $D_5 = 64^{-1}\Delta_5$ is the monic-versus-theta normalisation of the Gritsenko–Nikulin product; its square gives $\Delta_5^2/4096$. The minus sign belongs to the reciprocal OP/DT generating function, not to the Borcherds product. These scalar identities do not construct a compact critical CoHA, a quasi-NCCR realisation, a Hall–Drinfeld double, or a vanishing proof for* $\mathfrak{o}_{\mathrm{Hall}}$.

*Proof.* Let $D_5$ be the monic Borcherds product attached to the $\phi_{0,1}$-lift in the standard Gritsenko–Nikulin sign convention. The theta-normalised form has leading coefficient $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$, while $D_5$ is monic. Hence $D_5 = 64^{-1}\Delta_5$, and

$$D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2.$$

The physical unnormalised convention keeps the unscaled theta-constant normalisation and therefore writes $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. The Oberdieck–Pandharipande scalar convention uses the monic product, hence $\Phi_{10}^{\mathrm{OP}} = D_5^2$, and the reduced OP/DT generating function is the reciprocal with its separate sign convention: $-(\Phi_{10}^{\mathrm{OP}})^{-1} = -4096\Delta_5^{-2}$. This computation changes only the scalar normalisation. It supplies no orientation line, no compact extension correspondence, no monoidal vanishing-cycle realisation, no Hopf pairing, no radical quotient, no PBW filtration comparison, no quasi-NCCR realisation, no Hall–Drinfeld double, and no compact-CoHA construction. □

## 3. Complementarity

**Definition 9** (Principal-line scalar conventions)**.** Let $m, n \geq 1$, $m \neq n$, $M = \max(m, n)$, and $h = |m - n|$. The dimension-normalised principal even line is the scalar convention

$$c_{\mathrm{dim}}(k) = \frac{(M^2 - 1)k}{k + h}, \qquad \kappa_{\mathrm{ch}}^{\mathrm{dim}}(k) = \frac{1}{2} c_{\mathrm{dim}}(k).$$

The uniform principal-line convention is

$$c_{\mathrm{unif}}(k) = \frac{Mk}{k + h}, \qquad \kappa_{\mathrm{ch}}^{\mathrm{unif}}(k) = \frac{1}{2} c_{\mathrm{unif}}(k).$$

Both use the Sugawara-shift inversion $k^\vee = -k - 2h$. They are equal only in the accidental ranks solving $M = M^2 - 1$, and no such rank has $M \geq 2$.

**Proposition 10** (Complementarity constants)**.** *For the two scalar conventions above,*

$$\kappa_{\mathrm{ch}}^{\mathrm{dim}}(k) + \kappa_{\mathrm{ch}}^{\mathrm{dim}}(k^\vee) = M^2 - 1, \qquad \kappa_{\mathrm{ch}}^{\mathrm{unif}}(k) + \kappa_{\mathrm{ch}}^{\mathrm{unif}}(k^\vee) = M.$$

*Proof.* For any constant $C$ and $h \neq 0$,

$$\frac{1}{2} \frac{Ck}{k + h} + \frac{1}{2} \frac{C(-k - 2h)}{-k - h} = \frac{Ck + C(k + 2h)}{2(k + h)} = C.$$

Taking $C = M^2 - 1$ gives the dimension-normalised constant. Taking $C = M$ gives the uniform constant. □

*Remark* 11. The preceding proposition is an algebraic identity for two scalar conventions. It does not identify either scalar convention with the principal even line of $Y(\mathfrak{sl}(m|n))$. The rank-two computation below is proved in the stated normalisation; the family statement with constant $M^2 - 1$ is conjectural until the dimension-normalised sub-Sugawara central-charge formula and Verdier level-duality involution are constructed.

**Conjecture 12** (Dimension-normalised super-trace complementarity)**.** *The principal even sub-Sugawara line of $Y(\mathfrak{sl}(m|n))$ carries a Verdier-compatible dimension normalisation. In that normalisation,*

$$\kappa_{\mathrm{ch}}^{\mathrm{str}}\big(Y(\mathfrak{sl}(m|n))\big) + \kappa_{\mathrm{ch}}^{\mathrm{str}}\big(Y(\mathfrak{sl}(n|m))^!\big) = M^2 - 1.$$

*For $M \geq 3$ this requires a rank-uniform sub-Sugawara central-charge formula with numerator $M^2 - 1$ and a compatible Verdier level-duality involution. The uniform convention with numerator $M$ proves the different constant $M$.*

**Proposition 13** (Rank-two super-trace check)**.** *Assume the $Y(\mathfrak{sl}(2|1))$ principal even sub-Sugawara line has*

$$c_T(k) = \frac{3k}{k + 1}, \qquad \kappa_{\mathrm{ch}}^{\mathrm{str}}(k) = \frac{c_T(k)}{2},$$

*and Verdier level inversion $k^\vee = -k - 2$. Then*

$$\kappa_{\mathrm{ch}}^{\mathrm{str}}(k) + \kappa_{\mathrm{ch}}^{\mathrm{str}}(k^\vee) = 3 = \dim \mathfrak{sl}(2).$$

*Proof.* The inverted central charge is

$$c_T(-k - 2) = \frac{3(-k - 2)}{-k - 1} = \frac{3(k + 2)}{k + 1}.$$

Hence

$$\frac{1}{2} c_T(k) + \frac{1}{2} c_T(-k - 2) = \frac{3k + 3(k + 2)}{2(k + 1)} = 3.$$

The parity-swapped $Y(\mathfrak{sl}(1|2))$ line has the same scalar normalisation. □

**Definition 14** (Berezinian–Verdier datum). A Berezinian–Verdier datum for $Y(\mathfrak{sl}(m|n))$ is the tuple

$$\mathcal{V}_{m|n} = (\mathfrak{X}_{m|n}, \mathrm{Ber}_{\mathfrak{X}_{m|n}}, \delta_{m|n}, \mathbb{D}_{\mathrm{super}}, \langle -,- \rangle_{\mathrm{Ber}}, \iota_{m|n}, \lambda_{\mathrm{Ber}}).$$

Here $\mathfrak{X}_{m|n}$ is a chosen super moduli object carrying the Yangian-side $\mathcal{D}$-modules, $\mathrm{Ber}_{\mathfrak{X}_{m|n}}$ is its Berezinian dualising line, $\delta_{m|n} \in \mathbb{Z}$ is the cohomological shift included in the datum, and

$$\mathbb{D}_{\mathrm{super}}(\mathcal{M}) = \mathrm{RHom}(\mathcal{M}, \mathrm{Ber}_{\mathfrak{X}_{m|n}})[\delta_{m|n}]$$

is the Verdier functor. The pairing $\langle -,- \rangle_{\mathrm{Ber}}$ is non-degenerate on the principal line, $\iota_{m|n}$ is the level-duality involution, and $\lambda_{\mathrm{Ber}}$ is the scalar extraction rule defining $\kappa_{\mathrm{ch}}^{\mathrm{Ber}} = \lambda_{\mathrm{Ber}}(\langle -,- \rangle_{\mathrm{Ber}})$. Without this whole datum the symbol $\kappa_{\mathrm{ch}}^{\mathrm{Ber}}$ is undefined.

**Conjecture 15** (Berezinian normalisation). *Given a Berezinian–Verdier datum $\mathcal{V}_{m|n}$, the Berezinian complementarity conjecture is*

$$\kappa_{\mathrm{ch}}^{\mathrm{Ber}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\mathrm{ch}}^{\mathrm{Ber}}(Y(\mathfrak{sl}(n|m))^{!}) = |m-n|$$

*after the duality involution $\iota_{m|n}$, the pairing $\langle -,- \rangle_{\mathrm{Ber}}$, the shift $\delta_{m|n}$, and the scalar map $\lambda_{\mathrm{Ber}}$ are fixed. This is a conjecture about the Berezinian–Verdier datum. It is not a consequence of the super-Riccati recurrence and is independent of the super-trace constant $M^2-1$. A numerical value $|m-n|$ is therefore not defined until the datum $\mathcal{V}_{m|n}$ and the scalar extraction rule $\lambda_{\mathrm{Ber}}$ are part of the statement.*

*Remark* 16. The Verdier datum is the super-$\mathcal{D}$-module category or formal moduli object, the duality functor $\mathbb{D}_{\mathrm{super}}$, the Berezinian dualising line, the Berezinian pairing, the level-duality convention, and the scalar extraction rule. Any rank-wise compatibility with the super-Riccati coefficients is an additional comparison condition, not part of the recurrence.